

Unit 3.4 The Chain Rule.

Note Title

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Stewart, Section 3.4 p198-208.

- We are going to differentiate composite functions of the form: $f \circ g = f(g(x))$

The Chain Rule

If g is differentiable at x and f is differentiable at $g(x)$, then the composite function $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x and F' is given by the product

$$F'(x) = f'(g(x)) \cdot g'(x)$$

Example Find derivatives of

1] $\cos x$

2] $\sin \cos x$

3] $\cos 3x$

4] $\cos^3 x$

5] $\cos(x^3)$

The power rule combined with the Chain Rule

If n is any real number and $g(x)$ is differentiable, then

$$\frac{d}{dx} [g(x)]^n = n [g(x)]^{n-1} \cdot g'(x).$$

Example Differentiate

1] $\sqrt{t}$

2] $\sqrt{4 - 3t}$

3] $\tan \sqrt{t}$

4] $\sqrt{t \tan t}$

Example Differentiate

$$g(t) = \left(\frac{t-2}{2t+1} \right)^9$$

Suppose you have a composite function consisting of more than two functions

$F(x) = f(g(h(x)))$ where f, g & h are differentiable functions. Then the derivative is

$$F'(x) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x).$$

Example. Differentiate

$$1] \frac{d}{dx} e^x = e^x$$

$$2] \frac{d}{dx} e^{4x} = e^{4x} \cdot 4$$

$$3] \frac{d}{dx} \sin e^{2x} = \cos e^{2x} \cdot e^{2x} \cdot 2$$

$$4] \frac{d}{dx} (e^{\sin 4x}) = e^{\sin 4x} \cdot \cos 4x \cdot 4$$

$$5] \frac{d}{dx} (\sin e^{x^2}) = \cos e^{x^2} \cdot e^{x^2} \cdot 2x$$

Theorem

$$\frac{d}{dx} a^x = (\ln a) a^x \quad \text{for } a > 0$$

Proof.

$$e^{\ln a} = a \quad \text{for } a > 0$$

$$\frac{d}{dx} a^x = \frac{d}{dx} (e^{\ln a})^x$$

$$= \frac{d}{dx} e^{(\ln a)x}$$

$$= e^{(\ln a)x} \times \ln a$$

$$= a^x \times \ln a$$

Examples Find The derivative

1] 4^u

2] -4^u

3) $\operatorname{cosec} 4^u$

4) $4^{\operatorname{cosec} u}$

$$1) \frac{d}{du} 4^u = 4^u \cdot \ln 4$$

$$2) \frac{d}{du} -4^u = -4^u \ln 4$$

$$3) \frac{d}{du} \operatorname{cosec}(4^u)$$

$$= -\operatorname{cosec}(4^u) \cdot \cot(4^u) \cdot \ln 4 \cdot 4^u$$

$$4) \frac{d}{du} 4^{\operatorname{cosec}(3u)} = \ln 4 \cdot 4^{\operatorname{cosec}(3u)} \cdot (-\operatorname{cosec} 3u \cdot \cot 3u) \cdot 3$$

Example. Differentiate

$$1] \frac{d}{dx} [3ex^2 + 6\cos(\sin x)]$$

$$= 6ex - 6\sin(\sin x) \cdot \cos x$$

$$2) \frac{d}{dx} [3x^2 - 7x]^2$$

$$= 2(3x^2 - 7x)[6x - 7]$$

$$3) \frac{d}{dx} \left[\frac{e^{2x}}{3^x - \tan \sqrt{x}} \right]$$

$$= \frac{e^{2x} \cdot 2(3^x - \tan \sqrt{x}) - e^{2x} [\ln 3 \cdot 3^x - \sec^2 \sqrt{x} \cdot \frac{1}{2} x^{-\frac{1}{2}}]}{(3^x - \tan \sqrt{x})^2}$$

$$4) \frac{d}{dx} \left[\frac{4}{e^{2x} + 5^{3x}} \right]$$

$$= \frac{0 \cdot (e^{2x} + 5^{3x}) - 4 (e^{2x} \cdot 2 + \ln 5 \cdot 5^{3x} \cdot 3)}{(e^{2x} + 5^{3x})^2}$$

$$5) \frac{d}{dx} [3^{\sin 2x} \cot(3^{2x})]$$

$$= \frac{d}{dx} [3^{\sin 2x}] \cdot \cot(3^{2x}) + 3^{\sin 2x} \cdot \frac{d}{dx} [\cot(3^{2x})]$$

$$= \ln 3 \cdot 3^{\sin 2x} \cdot \cos 2x \cdot 2 \cdot \cot(3^{2x})$$

$$- 3^{\sin 2x} (\operatorname{cosec}^2 3^{2x}) \cdot \ln 3 \cdot 3^{2x} \cdot 2$$

Exercise. Find derivatives of The following functions

Do not simplify your answers.

1] $y = e^{2x}$

2] $f(x) = \sin e^{2x}$

3] $y = e^{\sin x^3}$

4] $f(t) = \frac{1}{t^4}$

5] $y = (t - t^3)^{10}$

6] $f(t) = \cos^4 3t$

$$7] f(x) = 2^x$$

$$8] f(x) = 2^{1-x}$$

$$9] f(x) = \cot(2^x)$$

$$10] f(t) = 7 \tan 3t - 8^{\tan t}$$

$$11] f(x) = \operatorname{cosec} \frac{\pi}{3}$$

$$12] f(x) = \frac{\sqrt{x^3 + 1}}{7 - \tan x}$$